

Microeconomic Analysis: Problem Set VII

Due to week 8: submit problems 1, 2 and 5

1 Problem: Cake Eating Problem

You have a cake which can be eaten during periods $t = 0, \dots, T$. Your utility from consuming share c_t of the cake on day t is $u = \sqrt{c_t}$, and discount factor equals to δ .

1. Suppose that $T = 2$. What is the optimal strategy of eating the cake?
2. Suppose that $T = \infty$.
 - (a) Write down Bellman Equation.
 - (b) Using your answer to finite horizon problem or otherwise suggest a functional form for the value function.
 - (c) Find the optimal cake-eating policy.
 - (d) What is your life-time utility from eating the cake? How does it depend on δ ?

2 Problem: Search.

There are three boxes each containing a prize drawn from a uniform distribution with support $[0, 1]$. Let $0 < s < 1/2$ be the cost of opening a single box.

1. Suppose that you have to decide how many boxes you want to open and then you can choose the largest prize. Describe the optimal search rule for various values of s .

2. Now suppose that you can open boxes sequentially, but cannot come back to previously opened boxes. Describe the optimal search rule for various values of s .
3. Now suppose that you can open boxes sequentially, but now you can get back to previously opened boxes. Describe the optimal search rule for various values of s .
4. For various values of s compare the expected number of sampled boxes in cases (1) and (3). For which s case (1) results in more search than (3)?

3 Problem: Search Engine

Consider a model of a search engine which matches consumers with firms. Firms sell differentiated product, which with probability θ yields valuation v uniformly distributed on $[0, 1]$ and with probability $1 - \theta$ gives match value of 0. Suppose that $\Pr(\theta = 1) = \alpha$ and $\Pr(\theta = 1/2) = 1 - \alpha$. When a consumer runs a search query, the search engine informs firms of their relevance and charges a click-through fee r . After that firms make decisions whether to join the search pool or not and set price for their product, p , and consumers search for the product at search cost s per firm.

1. Suppose that the search engine induces expected relevance θ^e . Derive consumer's optimal stopping rule.
2. For fixed θ^e what is the price charged by a firm of quality θ and its profit? How does the equilibrium price depend on search costs and average relevance? Explain the intuition behind your answer.
3. Express search engine's fee and profit in terms of marginal relevance θ^* , average relevance θ^e and s .
4. For which values of α the search engine prefers to distort the search pool? What r will it choose for various values of α ?

4 Problem: Exam 2021

Consider two boxes, each containing a sum of money drawn from the same continuous distribution $F(u)$ with support $[0, 1]$. You can open the first box for free. Then you can decide whether you want to open the second box or not. It costs s to open the second box. If you open it, you can either take the amount of money there or you can get back to box one, but it costs you b .

1. Assuming that s and b are sufficiently small, i.e. both search and return are possible, derive the optimal search rule.
2. How does the reservation value depend on s and b ?
3. For this question only assume that $F(u) = u$ and $s < 1/2$. For which values of b return never happens on the optimal search path?
4. Let $\bar{s}$ be the smallest level of search costs for which consumers never prefer opening the second box. How does $\bar{s}$ depend on b ?
5. Let v_1 be the reservation value if $(s, b) = (x, 0)$ and v_2 if $(s, b) = (0, x)$. How does v_1 relate to v_2 ? In which case there is more search and why?

5 Problem: Exam 2024

Pandora faces a 10,000 boxes she could open. Suppose that the distribution of prizes in box n is given by a probability distribution $F_n = 1 - e^{-u/n}$, where e is the base of natural logarithm, i.e. $\log e = 1$, $u \geq 0$ is a prize value and $n = 1, \dots, 10000$. Each box is clearly labelled as "box n ", so Pandora knows the distribution associated with each box. If Pandora decides not to open any of the boxes her utility is $u_0 = 0$. Suppose that Pandora can open box n at a cost s_n . Pandora may open boxes in any order she wants, and can stop at any moment and take the prize from the best opened box, or may decide not to open any boxes and take the outside option.

1. Find the expected value of prize in box n . How does it depend on n ?

2. Denote z_n the value for which Pandora is indifferent between accepting it immediately and opening box n only. Derive z_n .
3. Formulate the optimal search rule.
4. Suppose that $s_n = an$, $a > 0$. Show that if $a < 1$ Pandora wants to open box 10,000 first. What is the optimal search rule in this case? What Pandora will do if (i) $a > 1$, (ii) $a = 1$?
5. Suppose that $s_n = \frac{1}{e}n$, and suppose that the *actual* prize in box n equals to $u_n = \frac{10000}{n}$. Which box Pandora will end up choosing?